Non-Archimedean Projective Perspective: The Monna Map as a Visual Rendering Interface

Quni-Gudzinas, Rowan Brad

2026-08-16

Author: Quni-Gudzinas, Rowan Brad (QNFO Research Collective) **ORCID:** 0009-0002-4317-5604 **Date:** 2026-08-16 **Version:** v0.1 **WBS:** QNFO.UMP.009 · **Slug:** non-archimedean-projective-perspective **Status:** Published (Phase 5)

Abstract

Things farther away appear smaller, and two-point linear perspective is taken to approximate the eye. That entire phenomenology is Archimedean geometry. This paper asks what changes if the metric of physical space were instead non-Archimedean and ultrametric — would we perceive the same thing, and what would that mean ontologically? **Why a reader should care:** the question converts a familiar visual fact into a decisive constraint on a live research program (p-adic and ultrametric physics [1,6,7,8]) — smooth perspective does not rule out a discrete world, it only rules out *direct* perception of one, and the distinction marks the exact boundary between physics and phenomenology. **Premise depth:** the underdetermination result rests on two imported premises — (i) perception is a constructed interface, not a direct reading of the substrate (Kantian interface premise, adopted from the corpus's Unified Theory of Non-Archimedean Ontology [1]); (ii) the Monna map, a surjective non-continuous map from the p-adics to the reals, exists (established mathematics [9]). Given (i) and (ii) the conclusion follows at theorem level; without them it is unfounded. The naive alternative — ultrametric perception *without* a rendering — is already falsified by the observed smoothness of perspective, which is recorded here as falsified hypotheses C3–C4 rather than silently avoided.

---

1. Introduction and Positioning

The originating question (Obsidian note `_26228180341.md`, 2026-08-16): *"Things that are farther away appear smaller, and 2 point linear perspective is thought to approximate our eyes — but what if instead of an Archimedean linear geometry, perspective is actually* *non-Archimedean and ultrametric? Would we perceive the same thing, and what would that actually mean ontologically?"*

The QNFO corpus already treats ultrametric geometry as the candidate substrate for physics and computation [1,2,3,6,7,8], treats the observer as a node inside the ultrametric tree [4], and names the Monna map as the ratio-based interface between p-adic structure and continuous experience [5]. What the corpus lacks — identified by the full-corpus due-diligence sweep of 2026-08-16 (995 living-paper records, evidence in `artifacts/external-search/corpus-sweep-evidence.json`) — is the worked example of *visual projective perspective*: vanishing points, apparent-size scaling, and the line of sight, analyzed from both the Archimedean and the ultrametric side. This paper supplies that worked example in four moves: (2) the Archimedean anatomy of perspective; (3) the ultrametric demolition of its ingredients; (4) the rendering lemma that restores the appearances; (5) the underdetermination theorem and its ontological consequences.

---

2. The Archimedean Anatomy of Perspective [TERRITORY]

Linear perspective rests on three Archimedean ingredients.

(a) Similar triangles. For an object of height h at distance d from an eye, the apparent size is governed by the subtended angle $\theta \approx h/d$ for $d \gg h$. The proportionality is the similar-triangles theorem of Euclidean geometry, and it requires two things: a well-defined ratio of lengths, and a well-defined angle.

(b) Additive distance along geodesics. The distance d is accumulated along a straight line — a geodesic — and satisfies the Archimedean property: finitely many steps of any positive length can cross any finite gap. Distance is the sum of its parts, which is exactly what makes "farther" a continuous degree.

(c) Projective closure. Two-point perspective is projective geometry over $\mathbb{R}$: parallel lines are completed by ideal points — the vanishing points — on the line at infinity. The construction is algebraic over the reals and inherits the real field's order, completeness, and Archimedean valuation.

Every element of the familiar phenomenology — smooth shrinking, vanishing points, the horizon as a line — is Archimedean machinery.

---

3. The Ultrametric Demolition [TERRITORY]

Now suppose the metric is ultrametric: $d(x, z) \leq \max(d(x, y), d(y, z))$ for all x, y, z — the strong triangle inequality. Four standard consequences [BACKGROUND — not from search; standard metric theory] dismantle §2 piece by piece.

(a) Every triangle is isosceles (the two longest sides are equal). The geometry has no "betweenness" in the Euclidean sense; detours never add distance.

(b) Balls have no boundaries. In an ultrametric space every point of a ball is a center of that ball, and balls are clopen — simultaneously open and closed. A "surface" of a ball is empty. There is no horizon *line*: the boundary the vanishing point would sit on does not exist.

(c) The space is totally disconnected. No two points are joined by a continuous path: any continuous map $f\colon [0,1] \to X$ from the connected interval into a totally disconnected space is constant. There are no continuous light rays, no lines of sight, no geodesic spray along which "farther" could accumulate.

(d) Size, if imposed by fiat, is discrete. In the p-adic model, with $|\cdot|_p$, distances take values in $\{p^n : n \in \mathbb{Z}\}$. An apparent-size law $\theta \propto 1/d$ would then jump by factors of p — a digital staircase, not a smooth gradient.

So the direct answer to the originating question is: **no.** If perception were a direct reading of an ultrametric substrate, we would not perceive the same thing — we would perceive no continuous perspective at all, and the vanishing "point" would decompose into a tree of nested clopen cells.

But that is the naive answer, and it is already falsified by ordinary vision. The interesting question is not whether direct ultrametric perception survives — it does not — but whether a *rendered* one does. That is the next move.

---

4. The Rendering Lemma [MAP]

Definition (rendering interface). Let X be the substrate metric space and $V \cong \mathbb{R}^3$ the experienced visual field. A rendering is a surjective map $R\colon X \to V$. Perception measures only quantities defined in V.

The candidate map. The Monna map $M\colon \mathbb{Q}_p \to \mathbb{R}$ is the classical surjection from the p-adics onto the reals [9]; it is necessarily non-continuous, since $\mathbb{Q}_p$ is totally disconnected and $\mathbb{R}$ is connected. Pitkänen's "canonical identification" of p-adic and real physics is the same construction [10,11]. The corpus's Module 11 already proposes the Monna map as the ratio-based consciousness interface [5], and UNO states the interface premise directly: the mind "naturally smooths the discrete nature of reality into a continuous narrative" [1].

Lemma (rendering). Given premise (i) — perception is a rendering — the metric structure an observer experiences is the metric of the image space V, not the metric of X. In particular, a continuous-looking visual field is compatible with an ultrametric substrate whenever a surjective rendering exists. *Falsifiability C1: a no-go theorem ruling out surjective renderings of the required kind would kill the lemma.*

The Monna map is a *candidate*, not *the* map: it is non-injective and far from canonical (§8). What matters for the argument is existence, and existence is established [9].

---

5. The Underdetermination Theorem [MAP]

Theorem (perceptual underdetermination). Given premises (i) and (ii), no finite set of first-person visual observations — apparent sizes, parallax, vanishing-point convergence — can distinguish an Archimedean substrate from an ultrametric substrate.

Proof sketch. Both substrates render into the same image space V. For the Archimedean substrate, take R to be the familiar similarity rendering. For the ultrametric substrate, compose a surjection $\pi: X \to \mathbb{Q}_p$ with the Monna map, $R = M \circ \pi$. Every measurement the observer can make lives in V; the two renderings differ only behind the interface, where no measurement reaches. Therefore the observation records are identical. □

Corollary. "Would we perceive the same thing?" — yes, in principle: the same first-person world is compatible with both metrics.

The honest converse. This does not show the world is ultrametric. It shows the metric of the substrate is *not decidable from inside the interface*. The equivalence itself — not either disjunct — is the finding. And the naive converse fails too: smooth perspective does not establish an Archimedean world, because §4's rendering exists. Vision constrains the *interface*, not the world.

---

6. Ontological Consequences [MAP]

(1) The continuum is demoted to an appearance. Apparent continuity is evidence about the rendering, not about the substrate — Kant's forms-of-intuition thesis, sharpened by an explicit candidate geometry: the smoothness that makes linear perspective "look right" is the geometry we assumed before we drew.

(2) Distance becomes a degree of isolation, not a gap. In the corpus's informational ontology, substrate distance measures the degree of computational isolation between states [1]. "Far" is deep in the tree, not far along a line.

(3) Identity becomes tree-based. An object is a cluster in nested neighborhoods; individuation is partition-theoretic, not point-like.

(4) Ostrowski's theorem turns the metric into a postulate. The only nontrivial absolute values on $\mathbb{Q}$ are the real one and the p-adic ones [BACKGROUND — not from search; standard]. The Archimedean choice baked into every ruler, every eye, and every perspective diagram is therefore a genuine postulate, validated only by measurement — by how the physical universe responds — never by mathematics alone.

(5) The interface can be studied from inside. What we cannot decide, we can still probe: a rendering with structure leaks structure. The falsifiable probes are in §7.

---

7. Falsifiability Register

#	Claim	Type	Falsifiability condition	Status
C1	A surjective rendering from an	MAP	a no-go theorem rules out surjective renderings of the	OPEN

#	Claim	Type	Falsifiability condition	Status
	ultrametric model onto the visual field exists (rendering lemma)		required kind	
C2	First-person observations cannot separate Archimedean from ultrametric substrates (underdetermination)	MAP	a first-person observable invariant across all Archimedean renderings but absent from all ultrametric renderings is found	OPEN
C3	Direct (unrendered) ultrametric perception predicts discrete apparent-size jumps	naive model	smooth size–distance laws observed	**FALSIFIED** (ordinary vision); motivates §4
C4	Direct ultrametric perception predicts the absence of smooth parallax	naive model	smooth parallax observed	**FALSIFIED** (ordinary vision); motivates §4
C5	The interface, being a rendering of something, exhibits glitch-level discreteness	MAP-speculative	a worked Monna-map rendering (planned demo) reproduces all known smooth vision with no distinctive residue	OPEN, deferred to the P4 demo

---

8. Mandatory Symmetry Template

Where External Literature Supports the Claim

- **Dragovich (2003) [6], Zúñiga-Galindo (2023) [7], Zúñiga-Galindo & Mayes (2024) [8]** — p-adic quantum mechanics is a live, rigorous research program; the question "is the substrate non-Archimedean?" is not idle speculation, and [7] already shows p-adic models produce physical predictions (Planck-scale Einstein-causality violation).
- **Weiß (2024) [9]** — the Monna map is current mathematics, used to transfer real sequences with Poissonian pair correlations to the p-adic setting; the rendering map is a real object, not a metaphor.
- **UNO [1]** — the interface premise (the mind smooths the discrete into a continuous narrative) and the informational ontology of distance are stated corpus doctrine.

Where External Literature Constrains or Contradicts the Claim

- **No external paper applies the Monna map to visual perception.** The rendering lemma (§4) is this paper's extrapolation; the corpus's M11 [5] proposes it internally but is not external validation. A reader should treat §4–§6 as interpretive proposal, not established result.

- **The p-adic program treats $\mathbb{Q}_p$ as quantum position space at the Planck scale** [6,7,8] — not as the metric of macroscopic visual space. The perceptual application here is a further step the cited literature does not take.
- **The Monna map is non-injective and non-canonical.** Many surjections onto $\mathbb{R}$ exist; the theorem needs only *one*, so its force survives, but the map cannot be recovered from the phenomenology — which is exactly the underdetermination conclusion.
- **The theorem's force comes entirely from premise (i),** which is philosophical. No experiment proposed in §7 distinguishes the interfaces; C5 is the only probe that bites, and it is deferred. [NO CONSTRAINING EVIDENCE FOUND] applies to the naive-model falsifications C3–C4: nothing in the literature rescues direct ultrametric perception.

---

9. Conclusion

Naive ultrametric perception is falsified by smooth vision — recorded, not avoided (C3–C4). Rendered ultrametric perception is indistinguishable from Archimedean perception (§5). Together: the world's metric is a postulate, validated only by interaction, and the only thing perception ever measures is the interface. The originating question's answer is therefore both — we would perceive the same thing, and the meaning of that fact is that the continuum is a rendering, not a fact about the rendered.

Declarations

Funding. This work received no external funding. **Competing interests.** The author declares no competing interests. **Data availability.** No experimental data were generated. Evidence files for the due-diligence sweep and live API verifications are deposited (`artifacts/external-search/`). **Code availability.** No code beyond the citation-verification scripts (deposited) was required. **Ethics approval.** Not applicable. **Preprint policy.** This manuscript is posted as a preprint; it has not been submitted to any journal.

References

1. QNFO Research Collective. *A Unified Theory of Non-Archimedean Ontology*. Zenodo, 2026. DOI 10.5281/zenodo.19040000.
2. QNFO Research Collective. *Ultrametric Intelligence: A Non-Archimedean Foundation for Artificial General Intelligence*. Zenodo, 2026. DOI 10.5281/zenodo.19925320.
3. QNFO Research Collective. *Ultrametric Cognition*. Zenodo, 2026. DOI 10.5281/zenodo.19884971.
4. QNFO Research Collective. *The Observer Inside the Tree: Can Self-Location in an Ultrametric Structure Resolve the Inside/Outside Schism?* Zenodo, 2026. DOI 10.5281/zenodo.21473899.
5. QNFO Research Collective. *ULTRAMETRIC PHYSICS: Module 11: Monna Map as Ratio-Based Consciousness Interface*. Zenodo, 2026. DOI 10.5281/zenodo.19438889.
6. B. Dragovich. *p-Adic and Adelic Quantum Mechanics*. arXiv:hep-th/0312046, 2003.

7. W. A. Zúñiga-Galindo. *p-Adic Quantum Mechanics, the Dirac Equation, and the violation of Einstein causality*. arXiv:2312.02744, 2023.
8. W. A. Zúñiga-Galindo and N. P. Mayes. *p-Adic quantum mechanics, infinite potential wells, and continuous-time quantum walks*. arXiv:2410.13048, 2024.
9. C. Weiß. *P-adic Poissonian Pair Correlations via the Monna Map*. arXiv:2406.13255, 2024.
10. M. Pitkänen. *p-Adic description of Higgs mechanism I: p-Adic square root and p-adic light cone*. arXiv:hep-th/9410058, 1994.
11. M. Pitkänen. *p-Adic TGD: Mathematical Ideas*. arXiv:hep-th/9506097, 1995.